

Jessica Sun

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EDUCATION

McGill University

Aug. 2023 – Dec. 2027

Bachelor of Science in Computer Science - AI

Cumulative GPA: 3.79/4.00

- James McGill Scholarship
- Data Structures & Algorithms, Artificial Intelligence, Databases, Statistical Learning, Computer Systems
- Vice President of Finance of HKSN, an East Asian culture club

EXPERIENCE

Software Engineer Intern

Aug. 2026 - Present

Autodesk

Montreal, Canada

- Developed and shipped features and fixes to production across a large **Node/TypeScript monorepo**, spanning **React frontend, GraphQL backend**, and federated reporting services
- Fixed cache-consistency, filtering, and export defects across the reporting product
- Built and evaluated **LLM-powered report generation in AI insights**, using **Langfuse tracing and prompt evaluations** to measure output quality, identify failure modes, and refine prompts

Software Developer Intern

May. 2026 – Aug. 2026

Bell Canada

Montreal, Canada

- Designed and deployed an **offline LLM platform** on an NVIDIA DGX server using Qwen3.6 locally via Ollama, delivering secure, 100% on-premises inference for **130+ users**
- Built a **high-performance RAG pipeline** using HuggingFace embeddings and GPU-accelerated FAISS vector databases for instant semantic search, optimizing GPU execution to achieve **~100 tokens/sec inference speeds**
- Integrated a **secure Slack bot via Socket Mode** with dynamic whitelist access control, real-time response streaming, and an asynchronous SQLite/Streamlit telemetry dashboard tracking cost avoidance and performance metrics

Undergraduate Course Assistant

Sep. 2025 – Dec. 2025

McGill University Department of Mathematics and Statistics

Montreal, Canada

- Evaluated and graded assignments and exams on topics including **data management, visualization, exploratory analysis, and statistical computing methods in R**
- **Reviewed student code** for accuracy, clarity, and reproducibility
- Ensured consistent and fair assessment across a large undergraduate course in **statistics and computing**

Database Development Intern

Jun. 2022 – Aug. 2023

APLUS Learning Center

Rockville, Maryland

- Developed a Microsoft Access database to manage student records, incorporating **SQL, UI design, and entity-relationship modeling**
- Applied **SDLC principles** to deliver a reliable, user-friendly system with data validation and loss prevention

PROJECTS

Hooked | React, TypeScript, Chrome Extensions API

- Built a **Manifest V3 Chrome extension** that detects 100+ crochet terms on any page and shows hover definitions via a Shadow DOM tooltip, with US/UK terminology mapping
- Engineered **content-script DOM scanning** with TreeWalker traversal, chunked processing, and a debounced MutationObserver to highlight terms in **dynamically loaded content** without page reloads
- Synced popup, side-panel library, and service-worker contexts via typed message passing and a **custom Zustand adapter over chrome.storage.local**

Ataxx AI Game Agent | Python

- Worked as a team to engineer an **autonomous game-playing AI agent** in Python for the 2-player strategy game Ataxx using **Minimax search and Alpha-Beta pruning** to evaluate dynamic board positions
- Implemented **custom board-control heuristics** and move-ordering strategies to prune search trees efficiently and make competitive decisions under strict **real-time execution constraints**

McMetro Campus Metro Planner | Java, JUnit

- Implemented **Disjoint Set** (Union-Find) with path compression and union-by-size for near-constant operations
- Applied **graph algorithms** (sparse adjacency, Kruskal MST) to optimize passenger flow and track selection
- Developed a **Trie-based search system** for efficient prefix lookups and a **greedy interval scheduler** for resource location

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, HTML/CSS, Bash, C, R, Assembler, OCaml, SQL, Cypher

Frameworks: JUnit, React, Node.js

Developer Tools: Git, VS Code, IntelliJ, Eclipse, JDBC, ArcGIS Pro

Databases: PostgreSQL, IBM DB2, Microsoft Access

Certifications: AWS Certified Cloud Practitioner (2025)